p-value

Definition

A **p-value** is the probability of obtaining the observed result (or one more extreme), **assuming the null hypothesis is true**.

Simple Definition

> **The p-value tells us how surprising our sample result is if the null hypothesis is true.**

Or even simpler:

> **The p-value measures the evidence against the null hypothesis.**

Example

Suppose a beverage company claims:

Average bottle fill = **500 ml**

You randomly test **100 bottles**.

Average fill = **503 ml**

Question:

Is the machine really filling more than 500 ml?

Or

Is this difference just due to random sampling?

Hypothesis

H_0:

Average fill = 500 ml

H_1:

Average fill ≠ 500 ml

Calculate the Z-score

Given:

Sample Mean = 503

Population Mean = 500

Population SD = 10

Sample Size = 100

Standard Error

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \mathbf{1}$$

$$\textbf{Z-score} = \frac{503 - 500}{1} = \mathbf{3}$$

We got a Z-score of 3.

But how do we know whether this is large or small?

We convert it into a probability.

That probability is called the p-value.

Where does the p-value come from?

The total area under the curve = **1 (100%)**.

The Z-table tells us the area to the **left** of any Z-score.

For

$Z = 3$

The Z-table says

$P(Z \le 3) = 0.99865$

Meaning

99.865% of values lie to the left of $Z = 3$.

Finding the p-value

The area to the right is

$1 - 0.99865 = 0.00135$

Since our hypothesis is

$H1: \mu \neq 500$

It is a **two-tailed test**, so we consider both tails:

$$\text{p-value}=2\times0.00135=0.0027$$

Decision

Given

$\alpha = 0.05$

Since

$$\mathbf{0.0027<0.05}$$

Reject H_0.

What Does the p-value Mean?

The p-value is the probability of observing our data (or something more extreme) if the null hypothesis is true.

Large p-value

Result is not surprising.

Random chance can explain it.

Keep H_0.

Small p-value

Result is very surprising.

Random chance is unlikely.

Reject H_0.

p-value	Decision
$p < 0.05$	Reject H_0
$p > 0.05$	Fail to Reject H_0

0.05 is simply a commonly accepted threshold introduced by statistician Ronald Fisher. It means we're willing to accept a **5% chance of making a Type I Error** (rejecting a true null hypothesis).

Other fields may use different values:

- 0.10 → Less strict
- 0.05 → Standard
- 0.01 → More strict